M.Tech Program

Advanced Industry Integrated Programs

Jointly offered by University and LTIMindTree

Final Review

AI and LLM Powered Personalized Lifestyle Risk Awareness System (Web App)

Anushka Khatua
(Roll No: 24167030)

Knowledge partner

LTIMindtree

partner

Implementation

L&T EduTech

INTRODUCTION

- Modern lifestyles often involve poor sleep, low physical activity, high screen time and other factors not good for healthy lifestyle.
- Such habits can lead to stress, fatigue, and long-term health risks.
- Most existing wellness tools provide generic advice without personalization.
- There is a need to understand individual lifestyle patterns using data.
- This project aims to provide personalized lifestyle risk awareness through data-driven insights.

OBJECTIVES

- Develop ML + DL models for risk prediction
- Integrate Large Language Model (LLM) for personalized explanations
- Build interactive web app with real-time analysis
- Provide interpretable and actionable lifestyle insights

MODULE NAME

- **Dataset Acquisition Module** – Collect lifestyle and activity datasets
- **Preprocessing & EDA Module** – Clean, normalize, and analyze data
- **ML/DL Risk Prediction Module** – Predict risk using ML models and ANN
- **LLM Insight Generation Module** – Convert predictions into explanations
- **Backend API Module** – Handle data flow and model integration
- **Frontend Web Interface Module** – User input and result display
- **Visualization & Analytics Module** – Show insights using charts

METHODOLOGY

STEP 1

Dataset Acquisition

Lifestyle and Human Activity datasets with behavioural features

STEP 2

Data Preprocessing

Missing value handling, Normalization, Encoding, Feature Scaling

STEP 3

Exploratory Data Analysis

Statistical Analysis, Distribution Study, Correlation Analysis, Pattern Visualization

STEP 4

Model Development

Baseline ML models + ANN for Lifestyle Risk Classification

STEP 5

Model Evaluation

Accuracy, Precision, Recall, F1-score, Confusion Matrix

STEP 6

Final Deployment

Web-based App with User Input and Risk Dashboard

DATASETS USED

- **Human Activity Recognition Dataset:**
  Human Activity Recognition with Smartphones
- **Fitbit Lifestyle Tracker Dataset:**
  FitBit Fitness Tracker Data
- **Daily and Sports Activities Dataset:**
  Daily and Sports Activities - UCI Machine Learning Repository
- **Description:** Provides behavioral, physical activity, lifestyle, and daily habit data to analyze patterns and develop lifestyle risk prediction models.

DATASET DETAILS

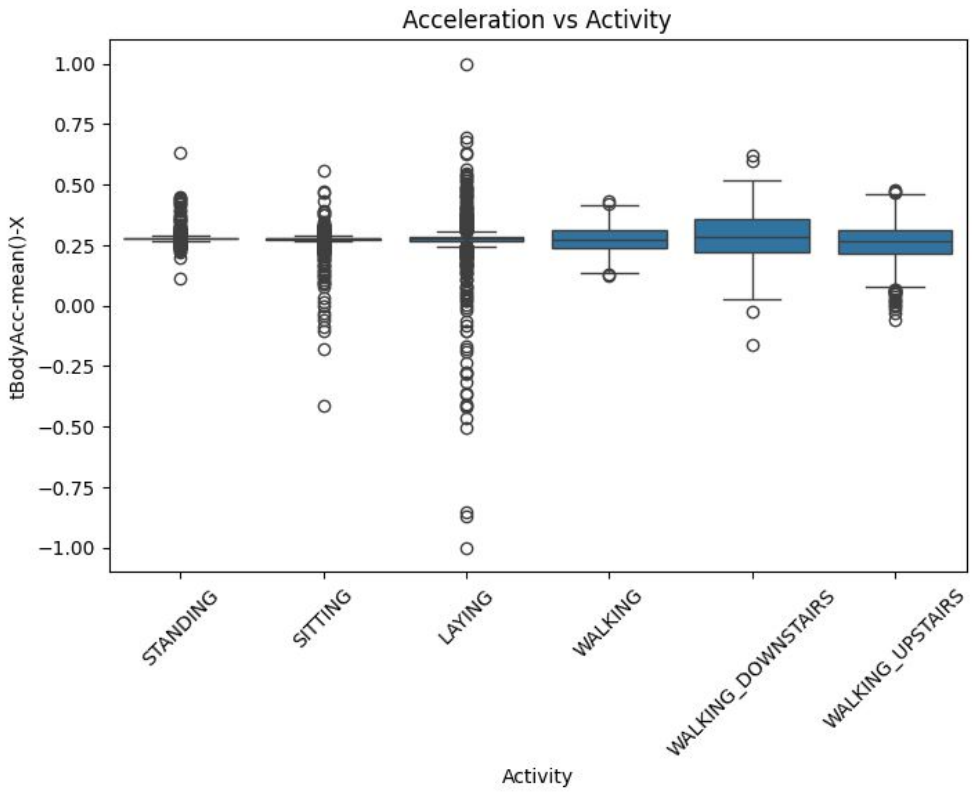
- **Combined Data Characteristics:**
  - **Total Samples** – Almost 10,000+ instances (combined)
  - **Subjects** – Multiple participants with diverse activity patterns
  - **Data Types** – Numeric, time-series, and sensor-based features
  - **Features** – Physical activity, movement signals, lifestyle habits
  - **Activities Covered** – Walking, sitting, standing, exercising, daily routines
  - **Feature Size** – High-dimensional (500+ features in HAR + additional lifestyle features)
  - **Integration** – Feature alignment and standardization across datasets
  - **Labels/Output** – Derived lifestyle risk levels (Low / Medium / High)

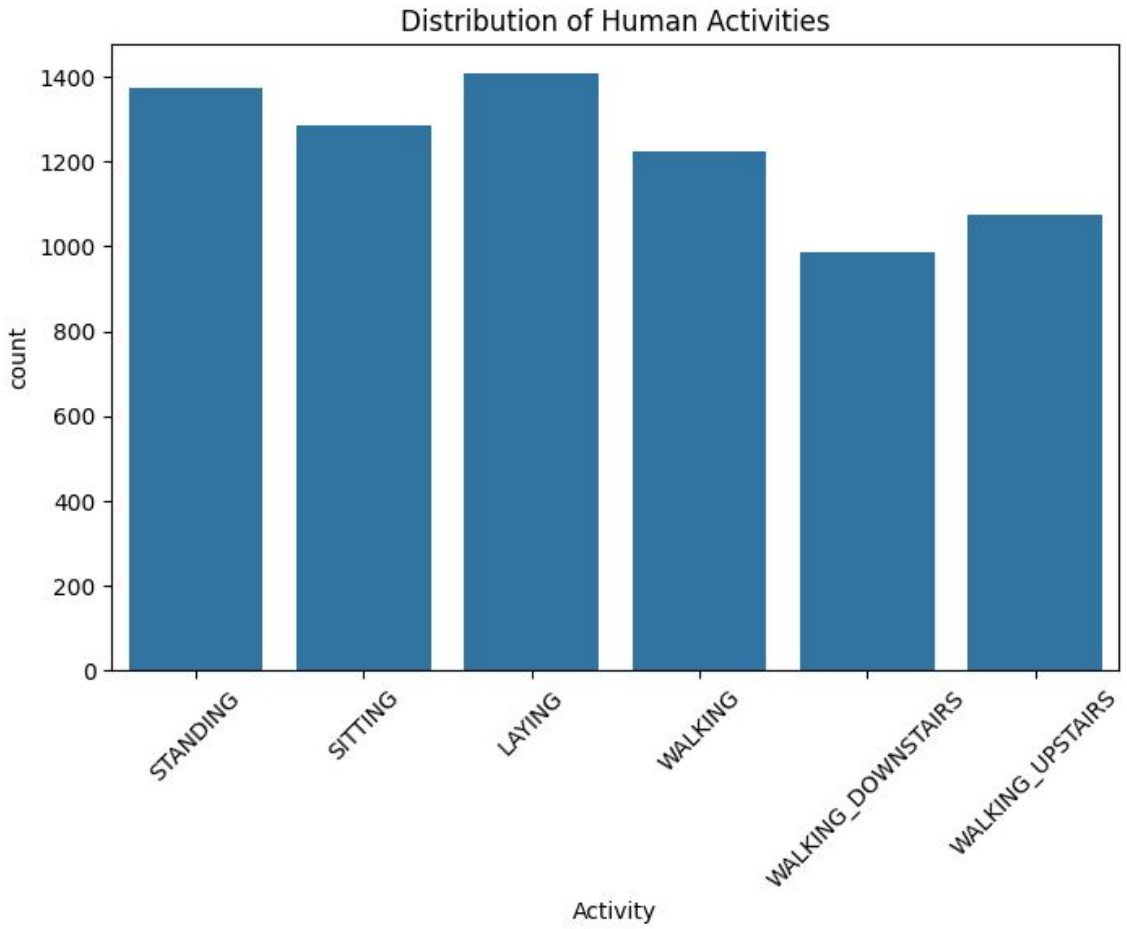
DATA PREPROCESSING & EDA

- Features are **normalized within range [-1, 1]**
- Mean values are centered → indicates **balanced data distribution**
- Standard deviation shows **controlled variability across features**
- No extreme outliers → data is **stable for model training**

	count	mean	std	min	25%	50%	75%	max
tBodyAcc-mean()-X	7352.0	0.274488	0.070261	-1.000000	0.262975	0.277193	0.288461	1.000000
tBodyAcc-mean()-Y	7352.0	-0.017695	0.040811	-1.000000	-0.024863	-0.017219	-0.010783	1.000000
tBodyAcc-mean()-Z	7352.0	-0.109141	0.056635	-1.000000	-0.120993	-0.108676	-0.097794	1.000000
tBodyAcc-std()-X	7352.0	-0.605438	0.448734	-1.000000	-0.992754	-0.946196	-0.242813	1.000000
tBodyAcc-std()-Y	7352.0	-0.510938	0.502645	-0.999873	-0.978129	-0.851897	-0.034231	0.916238
tBodyAcc-std()-Z	7352.0	-0.604754	0.418687	-1.000000	-0.980233	-0.859365	-0.262415	1.000000
tBodyAcc-mad()-X	7352.0	-0.630512	0.424073	-1.000000	-0.993591	-0.950709	-0.292680	1.000000
tBodyAcc-mad()-Y	7352.0	-0.526907	0.485942	-1.000000	-0.978162	-0.857328	-0.066701	0.967664
tBodyAcc-mad()-Z	7352.0	-0.606150	0.414122	-1.000000	-0.980251	-0.857143	-0.265671	1.000000
tBodyAcc-max()-X	7352.0	-0.468604	0.544547	-1.000000	-0.936219	-0.881637	-0.017129	1.000000

DATA PREPROCESSING & EDA





DATA PREPROCESSING & EDA

- Dataset contains **6 different activities** (walking, sitting, standing, etc.)
- Activity distribution is **fairly balanced**, hence no major class imbalance
- Dynamic activities (walking, stairs) show **higher variation in acceleration**
- Static activities (sitting, standing, laying) show **more stable values**
- Clear difference between active and inactive behaviors
- This separation helps the model **accurately classify activities**

ALGORITHMS/TOOLS

- **Algorithms:** Logistic Regression / Random Forest / Artificial Neural Network)
- **Loss Function:** Categorical Cross-Entropy
- **Optimizers:** Adam Optimizer
- **Evaluation Metrics:** Accuracy, Precision, Recall, F1-score, Confusion Matrix
- **Tools/Frameworks:** Python, Scikit-learn, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Seaborn, Google Colab, Large Language Model (LLM) integration
- **Web App Development & Deployment:** Hugging Face / API-based inference

SYSTEM PIPELINE

User Input → Backend API → ML/DL Model → Risk Score

↓

LLM Engine

↓

Personalized Insights

↓

Frontend Dashboard

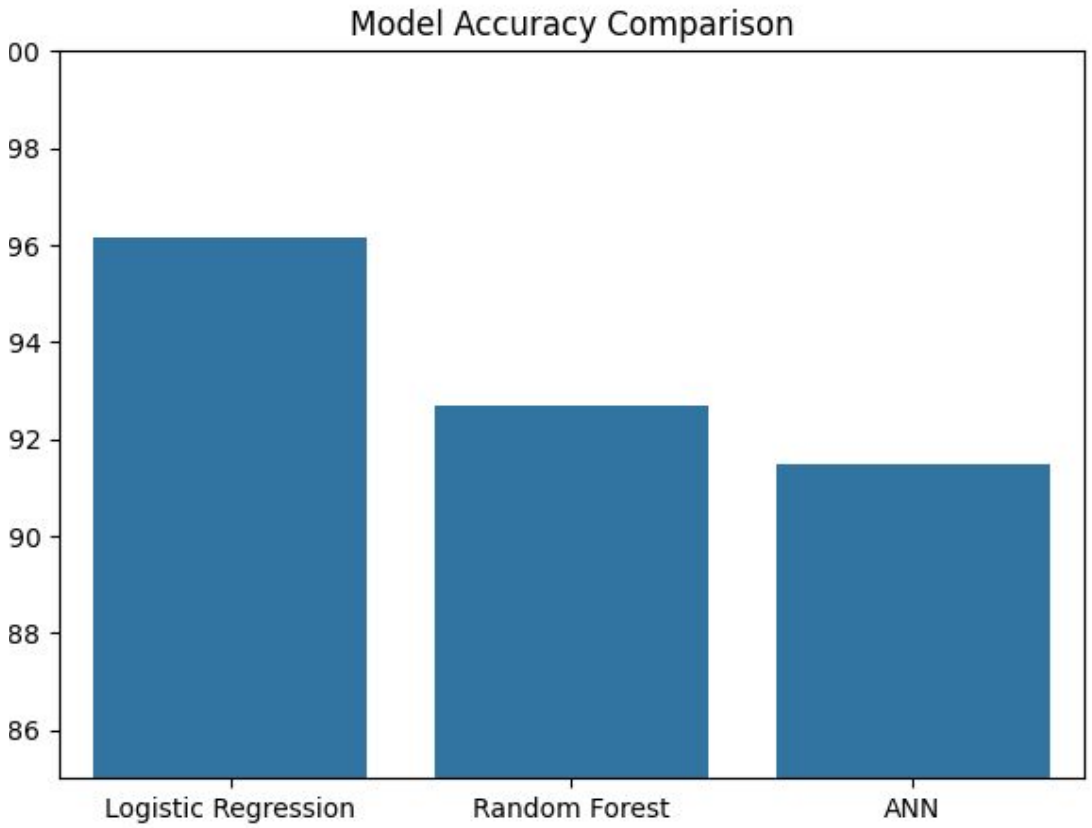
SYSTEM PIPELINE

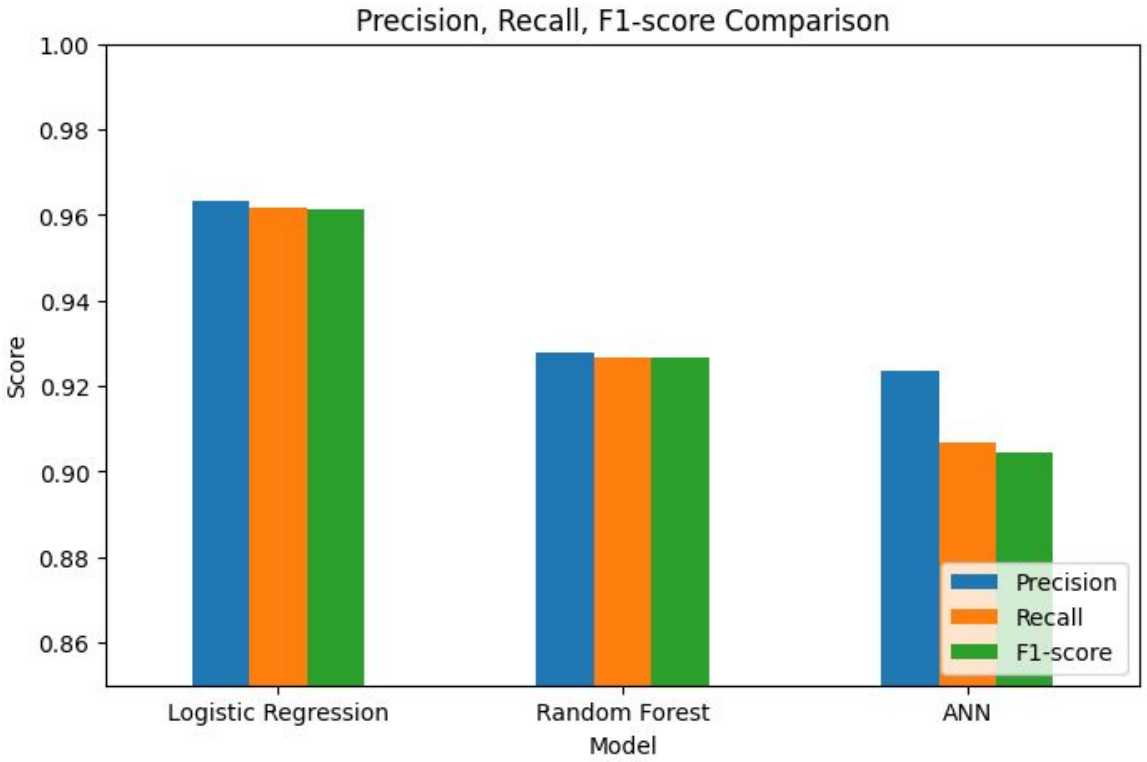
- **Description:**
  - User enters lifestyle data through web interface
  - Frontend sends input data to backend API
  - Backend processes and forwards data to ML/DL model
  - Model predicts lifestyle risk level (Low / Medium / High)
  - Prediction output is passed to LLM module
  - LLM generates personalized insights and recommendations
  - Backend sends results to frontend
  - Frontend displays risk score, analytics, and insights
  -

MODEL EVALUATION

- Evaluated using Accuracy, Precision, Recall, and F1-score
- Compared performance of all 3 models:
  - Logistic Regression **(96.16%)** , Random Forest **(92.67%)** , and ANN **(91.48%)**
- Logistic Regression achieved highest accuracy and better generalization
- Confusion Matrix used to analyze classification performance
- Model shows reliable prediction of lifestyle risk levels

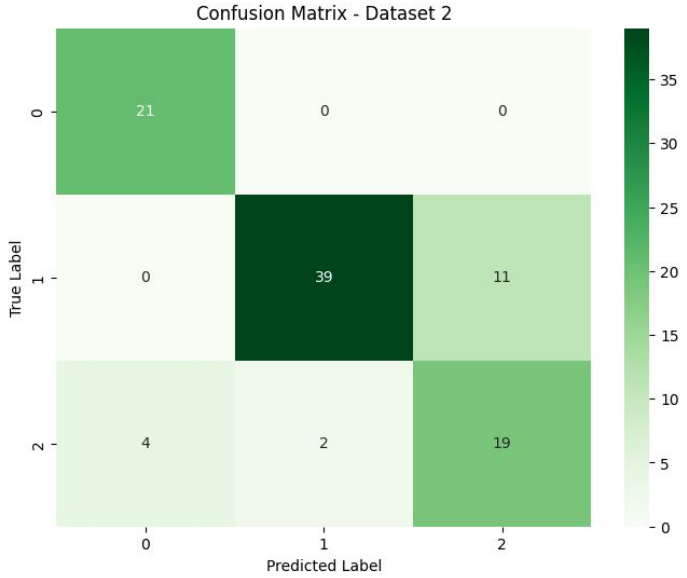
MODEL EVALUATION

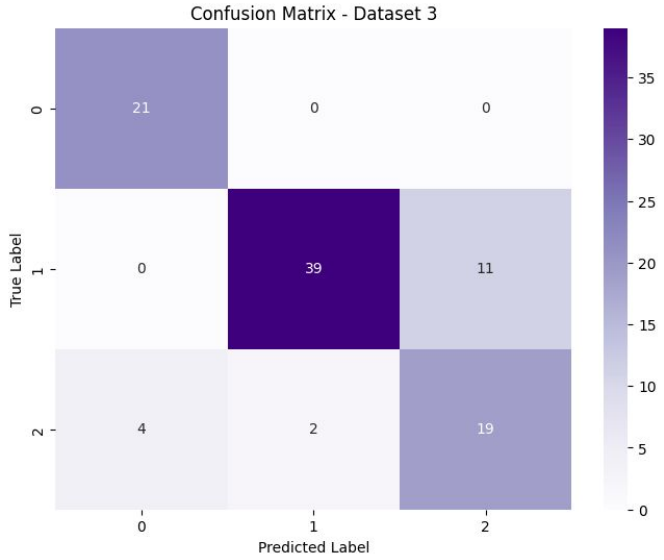




MULTI-DATASET VALIDATION

- Model evaluated and tested on additional Fitbit and Daily Activity datasets
- Consistent performance observed across datasets
- Demonstrates robustness and generalization ability
- Validates model effectiveness in real-world scenarios





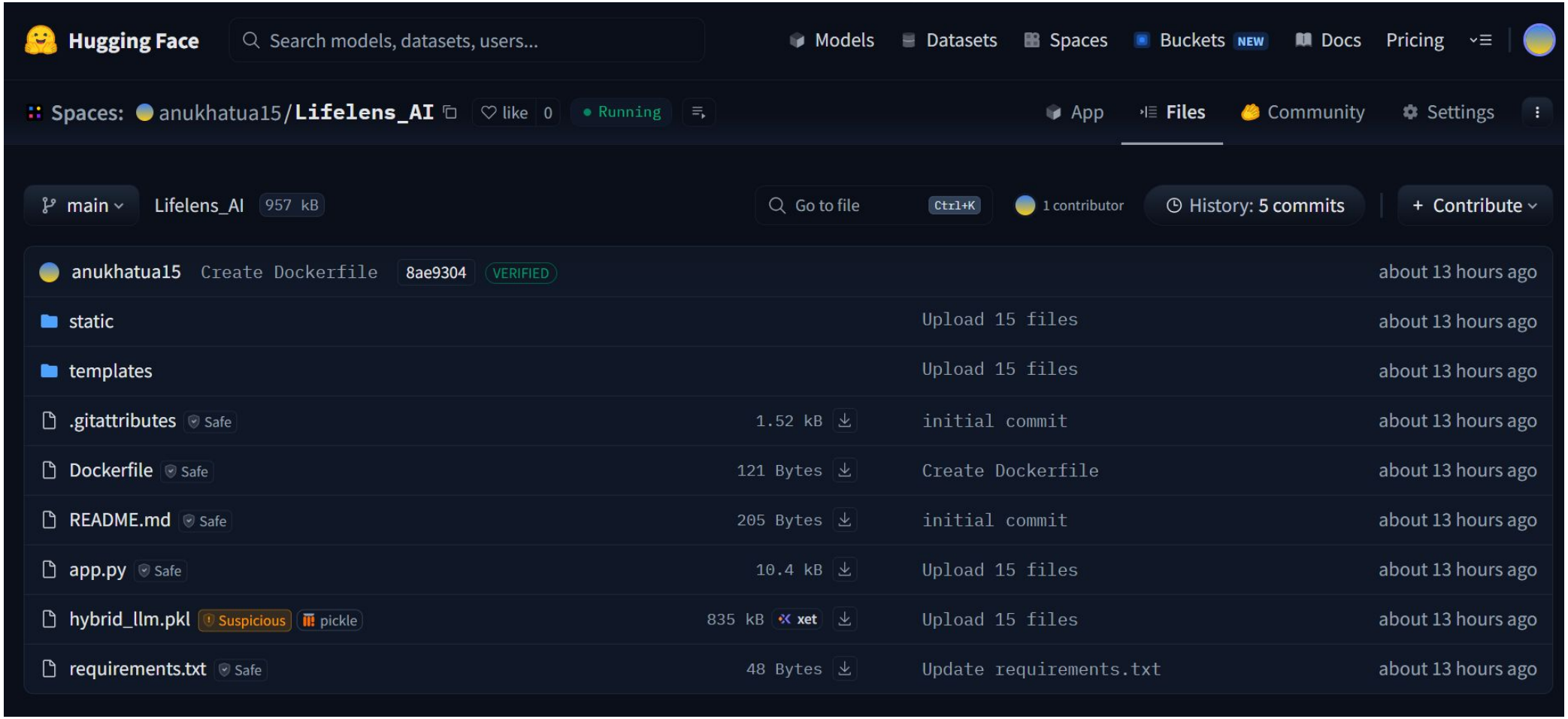
LLM INTEGRATION

- Outputs from multiple model components are combined and processed to form a unified interpretation input
- Rule-based logic with predefined conditions is applied to interpret model predictions in a structured manner
- Converted numerical prediction results into clear, human-readable explanations for better user understanding
- Generated personalized lifestyle recommendations based on risk patterns

DEPLOYMENT

- Model deployed using Flask/FastAPI backend
- Integrated ML model and LLM into backend pipeline
- Frontend connected through API calls
- Application deployed on Hugging Face platform
- Enables real-time prediction and user interaction

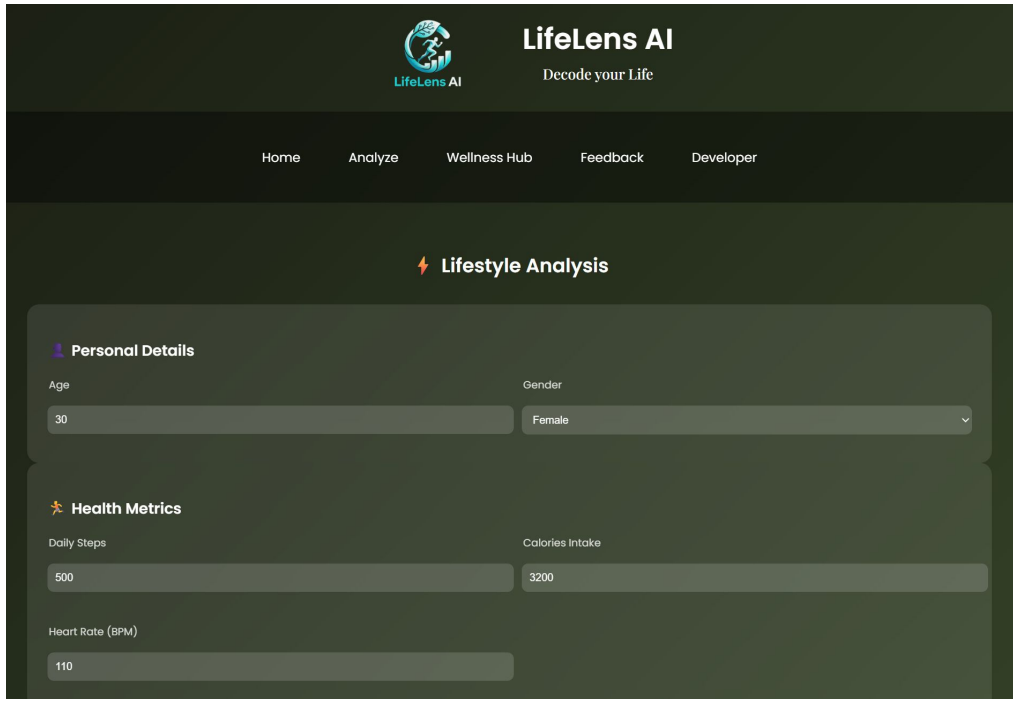
DEPLOYMENT

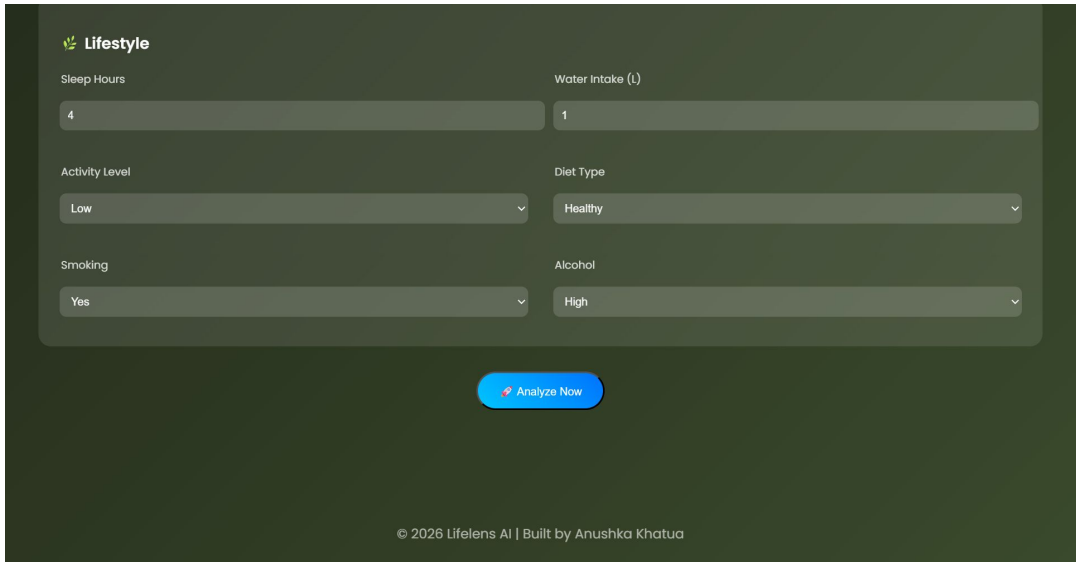

Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets NEW
Docs
Pricing
Spaces: anukhatua15/Lifelens_AI
like 0
Running
App
Files
Community
Settings
main
Lifelens_AI 957 kB
Go to file Ctrl+K
1 contributor
History: 5 commits
+ Contribute
anukhatua15 Create Dockerfile 8ae9304 VERIFIED about 13 hours ago
static Upload 15 files about 13 hours ago
templates Upload 15 files about 13 hours ago
.gitattributes Safe 1.52 kB initial commit about 13 hours ago
Dockerfile Safe 121 Bytes Create Dockerfile about 13 hours ago
README.md Safe 205 Bytes initial commit about 13 hours ago
app.py Safe 10.4 kB Upload 15 files about 13 hours ago
hybrid_llm.pkl Suspicious pickle 835 kB xet Upload 15 files about 13 hours ago
requirements.txt Safe 48 Bytes Update requirements.txt about 13 hours ago

APPLICATION DEMO

User Input Interface

- The user enters lifestyle-related data such as activity level, habits, and daily patterns through an interactive web form

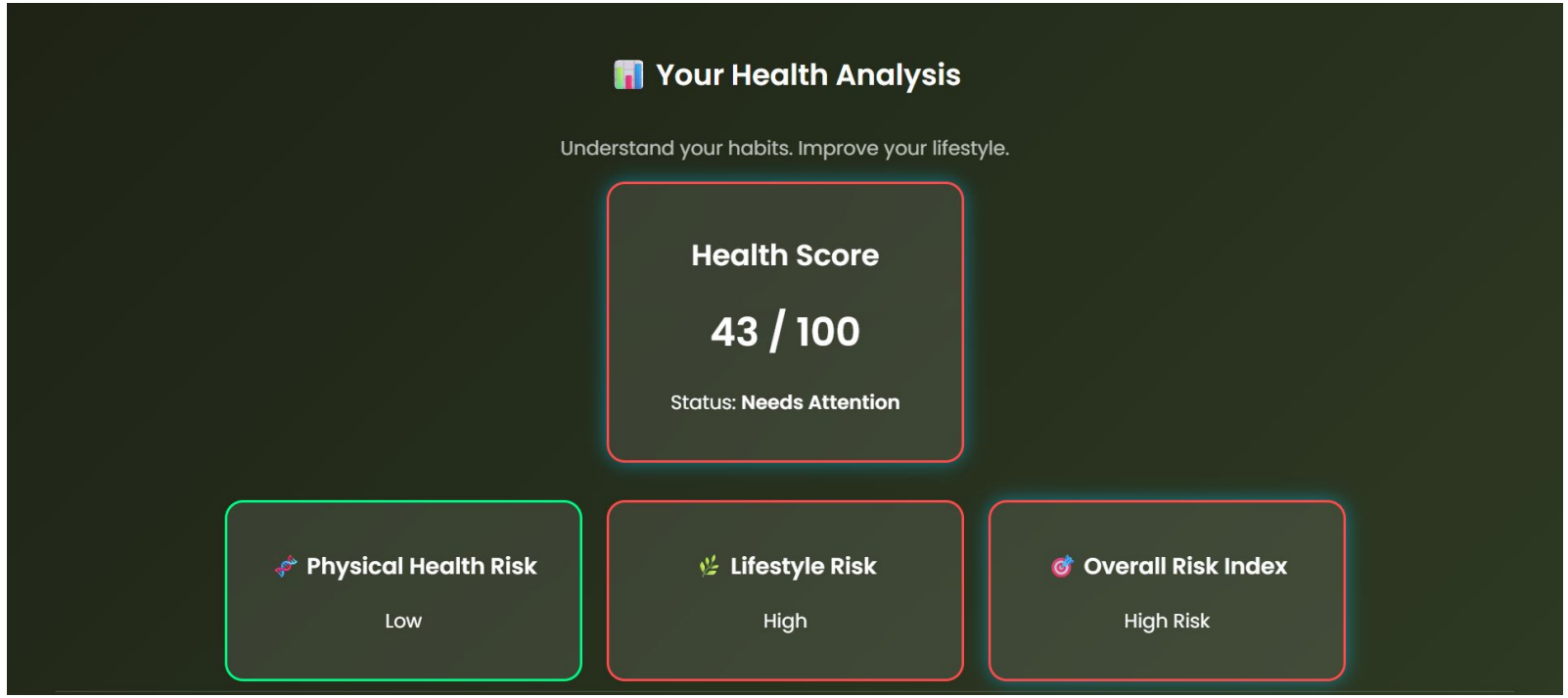




APPLICATION DEMO

Risk Prediction

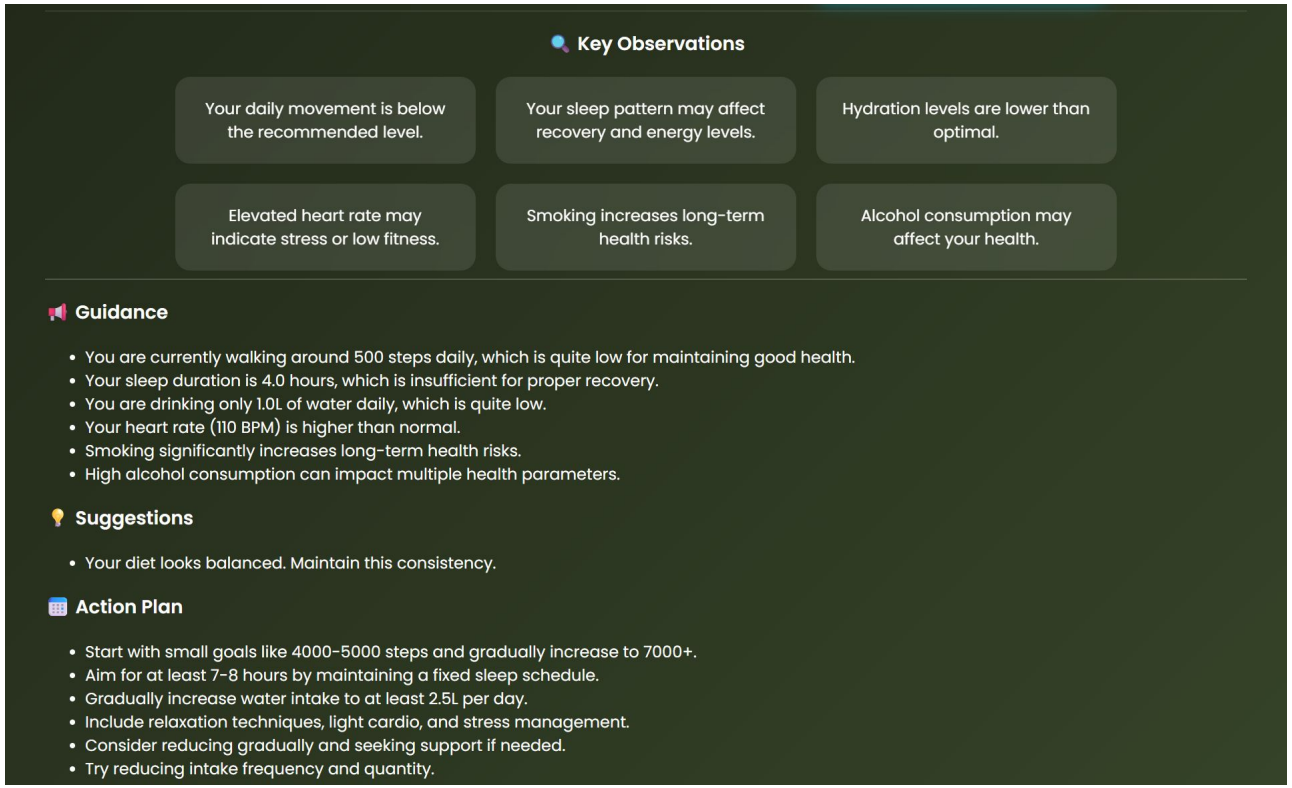
- The system processes the input data using the trained model and predicts the lifestyle risk level (Low / Medium / High) along with a health score



APPLICATION DEMO

Insight Generation

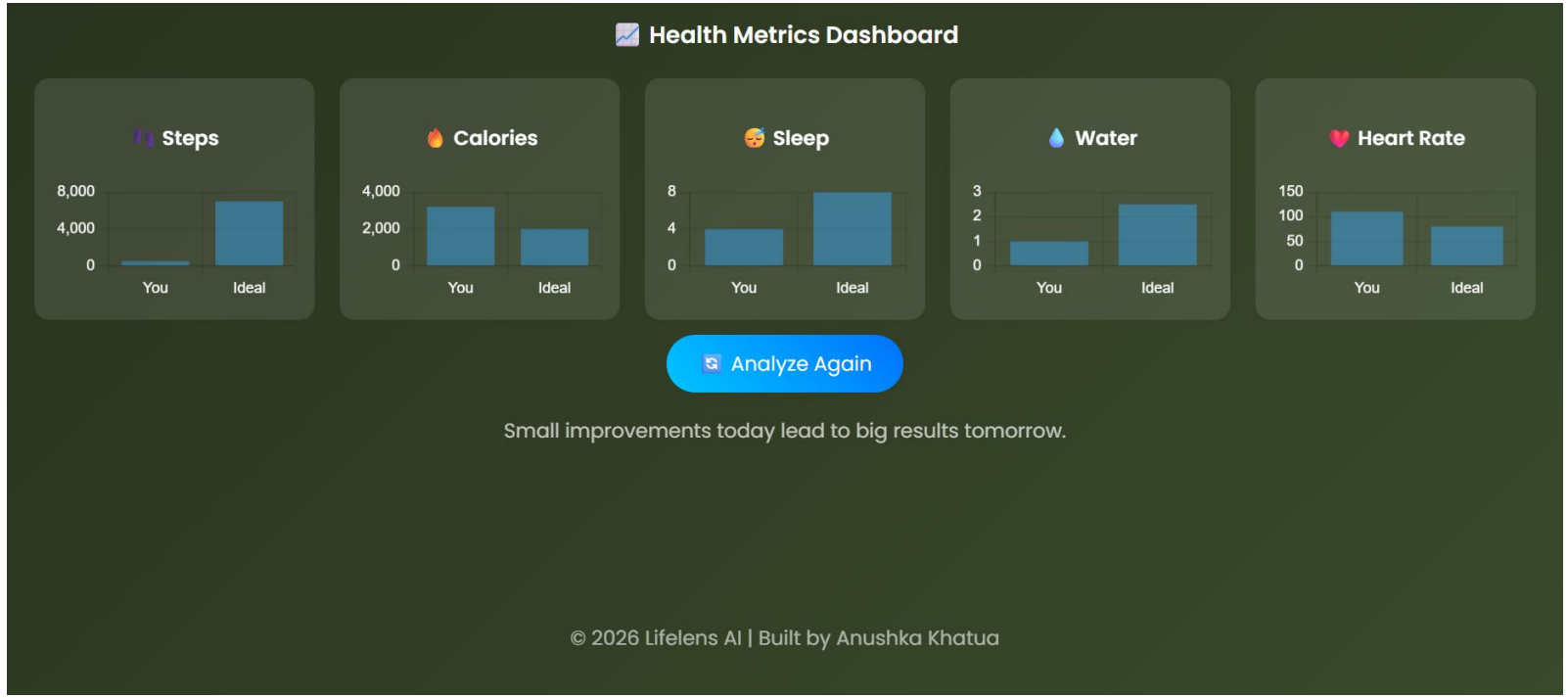
- Rule-based logic interprets the prediction results and generates clear, human-readable insights explaining the user’s lifestyle condition



APPLICATION DEMO

Visualization Dashboard

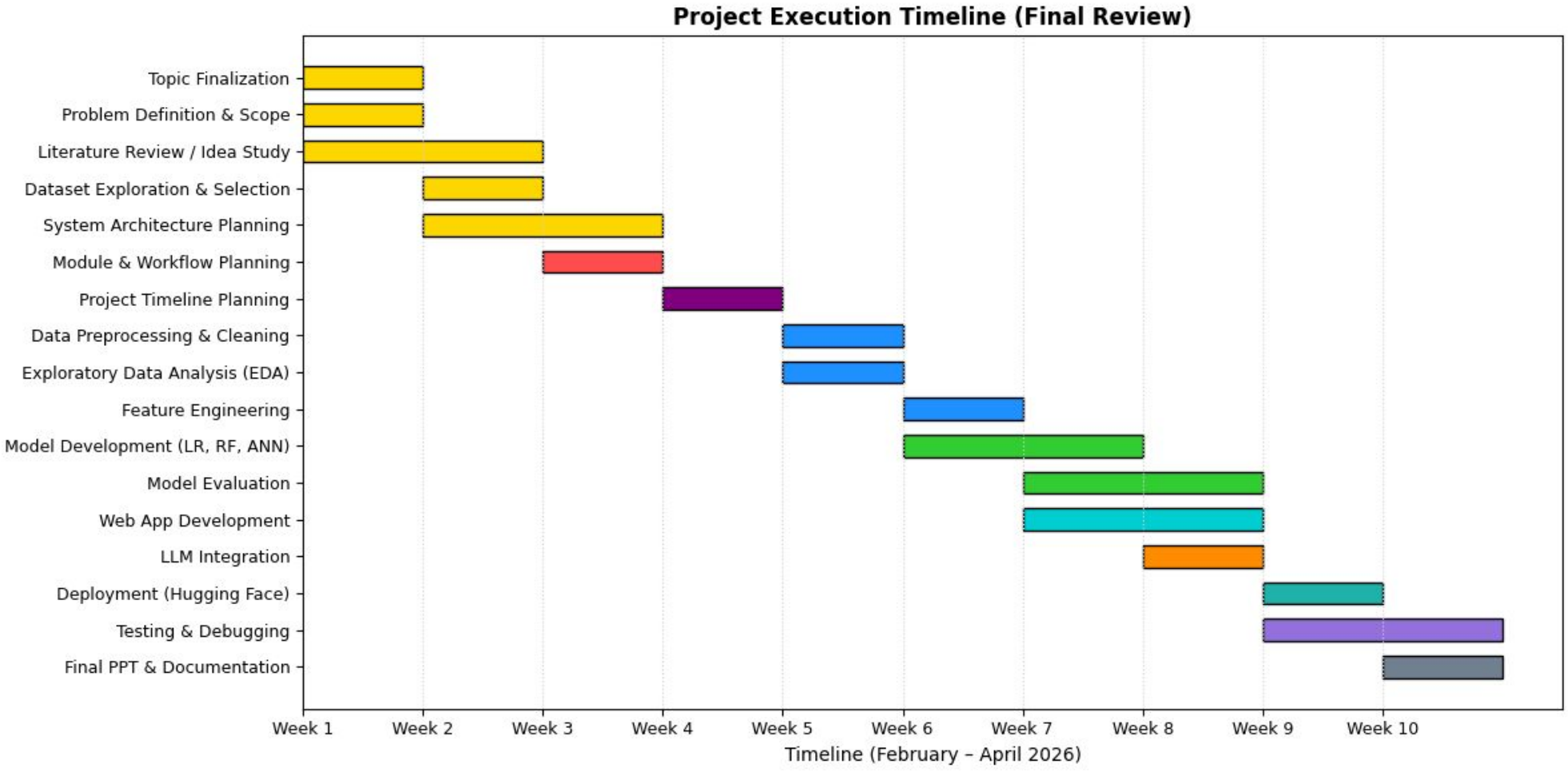
- The results are displayed through charts and graphs, helping users understand patterns and overall risk analysis effectively



CONCLUSION

- Developed an AI-based system for lifestyle risk prediction
- Compared ML/DL models and selected the best-performing model
- Validated model performance across multiple datasets
- Integrated LLM for personalized and interpretable insights
- Built and deployed a complete web-based application
- Provides a practical solution for lifestyle risk awareness

PROJECT PLAN/MILESTONES PROGRESS



REFERENCES

[1] UCI Machine Learning Repository, “Human Activity Recognition Using Smartphones Dataset,” Available: https://www.kaggle.com/datasets/uciml/human-activity-recognition-with-smartphones

[2] Arashnic, “Fitbit Fitness Tracker Data,” Kaggle Dataset, Available: https://www.kaggle.com/datasets/arashnic/fitbit

[3] UCI Machine Learning Repository, “Daily and Sports Activities Dataset,” Available: https://archive.ics.uci.edu/dataset/256/daily+and+sports+activities

REFERENCES

[4] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, 2011

[5] M. Abadi et al., “TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems,” 2015

[6] OpenAI, “Large Language Models (LLMs),” Available: https://openai.com

[7] Plotly Technologies Inc., “Plotly: Interactive Graphing Library,” 2015

[8] Hugging Face, “Model Deployment Platform,” https://huggingface.co

THANK YOU